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The N-soliton solutions to the Hirota and Maxwell–Bloch equation via the Riemann–Hilbert approach

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In this paper, we study the N-soliton solutions for the Hirota and Maxwell–Bloch equation with physical meaning. From the Lax pair and Volterra integral equations, the Riemann–Hilbert problem of this integrable equation is constructed. By solving the matrix Riemann–Hilbert problem with the condition of no reflecting, the N-soliton solutions for the Hirota and Maxwell–Bloch equation are obtained explicitly. Finally, we simulate the three-dimensional diagram of $|E|$ with 2-soliton solutions and the motion trajectory of t-axis in the case of different z.

Keywords: Hirota and Maxwell–Bloch equation; Riemann–Hilbert problem; N-soliton solutions; Lax pair; integrable equation.

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1. Introduction

In this paper, we consider the integrable Hirota and Maxwell–Bloch (HMB) equation with the zero boundary condition at infinity[1]

$$\begin{cases} E_z = iA_1(E_{tt} + |E|^2E) - A_2(E_{ttt} + 6|E|^2E_t) + A_3p, \\ p_t = 2i\omega p + 2E\eta, \\ \eta_t = -(Ep^* + E^*p), \end{cases} \tag{1.1}$$

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where A_1, A_2, A_3, ω are real constants, $*$ represents the complex conjugate, E is the field, p is the polarization and η is the population inversion. In fact, the HMB equation has several applications about higher-order dispersion, nonlinear effect, self-steepening, nonlinear fiber optics in physics, and the soliton solutions to the HMB equation are also obtained by Darboux transformation.[2] In fact, the soliton solutions of the HMB equation can be constructed by the matrix Riemann–Hilbert problem which was born out of inverse scattering transformation in 1997.[3] A few years later, people applied it to solve integrable equations, for example, the modified Korteweg–de Vries equation,[4] the multi-component AKNS integrable hierarchies,[5] the nonlocal Fokas–Lenells equation,[6] the Kundu-nonlinear Schrödinger equation[7] the N-coupled Hirota equations,[8] the coupled modified nonlinear Schrödinger equations,[9] the coupled nonlinear Schrödinger equations,[10] the coupled derivative Schrödinger equation,[11] the Kundu-Type equation[12] and the generalized Sasa–Satsuma equation[13,14] so on. Furthermore, the Riemann–Hilbert method can not only be applied to construct the N-soliton solutions, but also be used for the initial-boundary value problems and the long-time asymptotic behaviors of defocusing systems.[15–22]

This paper is arranged as follows. In Sec. 2, from the Lax pair of the HMB equation and the full derivative form (2.4), we obtain the asymptotic condition of ϕ. By introducing two related Volterra integral equations and some definitions, we can construct Riemann–Hilbert problem of the HMB equation. In Sec. 3, the N-soliton solutions to the HMB equation are obtained in detail and some three-dimensional images of N-soliton solutions are shown. In Sec. 4, we give a brief conclusion of this paper and present some discussions in integrable systems.

2. Lax Pair for the HMB Equation

From Ref. 2, we get that the HMB equation satisfies the Lax pair with $A_1 = 2, A_2 = -1, A_3 = 2$

$$\begin{cases} \Psi_t = U\Psi = (-i\lambda\sigma_3 + U_0)\Psi, \\ \Psi_z = V\Psi = \left(-4i\lambda^3\sigma_3 + \lambda^2 V_2 + \lambda V_1 + V_0 + \dfrac{i}{\lambda+\omega}V_{-1}\right)\Psi, \end{cases} \quad \Psi = (\Psi_1, \Psi_2), \tag{2.1}$$

where

$$U_0 = \begin{pmatrix} 0 & E \\ -E^* & 0 \end{pmatrix}, \quad V_2 = \begin{pmatrix} -2i & -4E \\ 4E^* & 2i \end{pmatrix},$$

$$V_1 = \begin{pmatrix} -2i|E|^2 & 2E - 2iE_t \\ -2E^* - 2iE_t^* & 2i|E|^2 \end{pmatrix},$$

$$V_0 = \begin{pmatrix} i|E|^2 - EE_t^* + E_tE^* & 2|E|^2E + iE_t + E_{tt} \\ -2|E|^2E^* + iE_t^* - E_{tt}^* & -i|E|^2 + EE_t^* - E_tE^* \end{pmatrix},$$

$$V_{-1} = \begin{pmatrix} \eta & -p \\ -p^* & -\eta \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the third Pauli's matrix, and $\lambda \in \mathbb{C}$ is the spectral parameter. In order to facilitate the calculation and analysis below, we introduce a matrix function $\phi(t, z; \lambda)$,

$$\phi = \Psi e^{i\lambda t\sigma_3 + 4i\lambda^3 z\sigma_3}, \quad z, t \to \pm\infty. \tag{2.2}$$

Then, we can get a simpler form of the Lax pair (2.1) as follows:

$$\begin{cases} \phi_t + i\lambda[\sigma_3, \phi] = U_0\phi, \\ \phi_z + 4i\lambda^3[\sigma_3, \phi] = \breve{V}\phi, \end{cases} \tag{2.3}$$

where $\breve{V} = \lambda^2 V_2 + \lambda V_1 + V_0 + \frac{i}{\lambda+\omega} V_{-1}$. Note that Eq. (2.3) can be rewritten in full derivative form as follows[7]:

$$d(e^{-i(\lambda t + 4\lambda^3 z)\hat{\sigma}_3}\phi) = \Upsilon\phi, \tag{2.4}$$

where

$$\Upsilon = e^{-i(\lambda t + 4\lambda^3 z)\hat{\sigma}_3}(U_0 dt + \breve{V} dz). \tag{2.5}$$

According to the matrix spectral problems of Lax pair (2.1), one can obtain the asymptotic condition

$$\Psi \sim e^{-i(\lambda t + 4\lambda^3 z)\sigma_3}, \quad t, z \to \pm\infty. \tag{2.6}$$

Based on the generalized Liouville's formula and the equivalent variable transformation, it shows that

$$\det \phi = 1. \tag{2.7}$$

In order to obtain the important scattering matrix, we can start from the analytic of the spectral function $\phi(t, z; \lambda)$. First, we introduce two related Volterra integral equations as follows:

$$\phi_L(t; \lambda) = I + \int_{-\infty}^{t} e^{-i\lambda(t-y)\hat{\sigma}_3}(U_0(y)\phi_L(y, \lambda))dy, \tag{2.8}$$

$$\phi_R(t; \lambda) = I - \int_{t}^{\infty} e^{-i\lambda(t-y)\hat{\sigma}_3}(U_0(y)\phi_R(y, \lambda))dy, \tag{2.9}$$

where I is an identity matrix. Let

$$\phi_L = (\phi_L^1, \phi_L^2) = \begin{pmatrix} \phi_L^{11} & \phi_L^{12} \\ \phi_L^{21} & \phi_L^{22} \end{pmatrix}, \tag{2.10}$$

$$\phi_R = (\phi_R^1, \phi_R^2) = \begin{pmatrix} \phi_R^{11} & \phi_R^{12} \\ \phi_R^{21} & \phi_R^{22} \end{pmatrix}, \tag{2.11}$$

where ϕ_L^1 and ϕ_R^1 are the first columns of ϕ_L and ϕ_R, ϕ_L^2 and ϕ_R^2 are the second columns of ϕ_L and ϕ_R, respectively. Based on the Volterra integral equation (2.8), it shows that

$$e^{-i\lambda(t-y)\sigma_3}U_0e^{i\lambda(t-y)\sigma_3} = \begin{pmatrix} 0 & Ee^{2i\lambda(t-y)} \\ -E^*e^{-2i\lambda(t-y)} & 0 \end{pmatrix}, \tag{2.12}$$

by calculating the formula $e^{-i\lambda(t-y)\sigma_3}U_0e^{i\lambda(t-y)\sigma_3}$, directly. From the calculation results and the defined Eqs. (2.10)–(2.11), it implies that ϕ_L^1 is analytic in $C^+ = \{\lambda \,|\, Im\lambda < 0\}$, ϕ_L^2 is analytic in $C^- = \{\lambda \,|\, Im\lambda > 0\}$, ϕ_R^1 is analytic in C^+, ϕ_R^2 is analytic in C^-, respectively.

Based on Abel's formula and the formula $tr(U_0) = 0$, it is easy to obtain that the asymptotic relations of ϕ_L and ϕ_R are independent of all t,

$$\begin{cases} \phi_L \to I, & t \to -\infty, \\ \phi_R \to I, & t \to \infty. \end{cases} \tag{2.13}$$

Considering two matrix Just solutions of the spectral problem (2.3), they are linearly related by the scattering condition $S(\lambda) = (s_{ij})_{2\times 2}$,

$$\phi_L E = \phi_R E S(\lambda), \quad E = e^{i\lambda t\sigma_3}. \tag{2.14}$$

It is worth noting that $\det S(\lambda) = 1$ because of $\det \phi_L = 1$ ($\det \phi_R = 1$), assume

$$\phi_L^{-1} = \begin{pmatrix} (\phi_L^1)^{-1} \\ (\phi_L^2)^{-1} \end{pmatrix}, \quad \phi_R^{-1} = \begin{pmatrix} (\phi_R^1)^{-1} \\ (\phi_R^2)^{-1} \end{pmatrix}, \tag{2.15}$$

this suggests that $(\phi_L^1)^{-1}$, $(\phi_R^2)^{-1}$ are analytic in C^+, $(\phi_L^2)^{-1}, (\phi_R^1)^{-1}$ are analytic in C^-. From the deformation of (2.14), it is obvious that the analytic of $S(\lambda)$ can be given as follows:

$$ES(\lambda)E^{-1} = \phi_R^{-1}\phi_L = \begin{pmatrix} (\phi_R^1)^{-1}\phi_L^1 & (\phi_R^1)^{-1}\phi_L^2 \\ (\phi_R^2)^{-1}\phi_L^1 & (\phi_R^2)^{-1}\phi_L^2 \end{pmatrix}, \tag{2.16}$$

then s_{11} is analytic in C^-, the other elements in $S(\lambda)$ are analytic in C^+. The purpose of the previous preparation and analysis is to construct the Riemann–Hilbert problem for the HMB equation, once the Riemann–Hilbert problem of the HMB equation is constructed, its corresponding solution can be given. First, we introduce two matrix functions

$$\begin{cases} R_R = (\Phi_L^1, \Phi_R^2) = \Phi_L H_1 + \Phi_R H_2, \\ R_L = ((\Phi_L^1)^{-1}, (\Phi_R^2)^{-1}) = H_1(\Phi_L)^{-1} + H_2(\Phi_R)^{-1}, \end{cases} \tag{2.17}$$

where

$$H_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{2.18}$$

$$H_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2.19}$$

According to the analytic of Volterra integral equations Φ_L, Φ_R, it is obvious that R_R is analytic in C^-, and satisfies the asymptotic relation $R_R \to I$ as $\lambda \in C^- \to \infty$. Similarly, R_L is analytic in C^+, where $R_L \to I$ as $\lambda \in C^+ \to \infty$, and from Eq. (2.14), we obtain

$$E^{-1}\phi_L^{-1} = S^{-1}(\lambda)E^{-1}\phi_R^{-1}, \tag{2.20}$$

where $S^{-1}(\lambda) = (\breve{s}_{ij})_{2\times 2}$.

For the HMB equation (1.1), the relevant matrix Riemann–Hilbert problem can be proposed as follows:

(1) R_R is analytic in C^-, R_L is analytic in C^+, respectively,

(2) $\begin{cases} R_R \to I, \lambda \in C^- \to \infty, \\ R_L \to I, \lambda \in C^+ \to \infty, \end{cases}$

(3) $G(t, z; \lambda) = R_L R_R$,

where

$$G(t, z; \lambda) = \begin{pmatrix} 1 & \breve{s}_{12}e^{-2i\lambda t} \\ s_{21}e^{2i\lambda t} & 1 \end{pmatrix} \tag{2.21}$$

is the jump matrix. Note that $\breve{s}_{11}s_{11} + \breve{s}_{12}s_{21} = 1$ because of the $\det S(\lambda) = 1$.

3. *N*-Soliton Solutions for the HMB Equation

In this section, the N-soliton solutions to the HMB equation are constructed by the Riemann–Hilbert problem proposed above. Now, let us assume the Riemann–Hilbert problem (2.21) is irregular, it implies that $\det R_R, \det R_L$ are meaningful in their own analytic domains except for some zeros. According to the definitions of R_R, R_L (2.17) and the scattering relationship (2.14), we find that

$$\det R_L = \breve{s}_{11}(\lambda), \quad \det R_R = s_{11}(\lambda), \tag{3.1}$$

it shows that of $\det R_L$ and $\det R_R$ possess the same zeros as $\breve{s}_{11}(\lambda)$ and $s_{11}(\lambda)$, respectively. Moreover, one can deduce that $R_{L,R}^{\ddagger}(\lambda^*) = R_{L,R}^{-1}(\lambda)$, where the superscript $\ddagger$ represents conjugate transpose, due to the spectral matrix U_0 is anti-Hermitian ($U_0^{\ddagger} = U_0^{-1}$), and the formulas

$$S^{\ddagger}(\lambda^*) = S^{-1}(\lambda), \quad \Phi_R^{\ddagger}(\lambda^*) = \Phi_L^{-1}(\lambda), \quad \lambda \in C^+, \tag{3.2}$$

are easy to obtain. From the relation of $S^{\ddagger}(\lambda^*) = S^{-1}(\lambda)$, we further have $s_{11}^*(\lambda^*) = \breve{s}_{11}^{-1}(\lambda)$. Let $\lambda_1, \lambda_2, \ldots, \lambda_N \in C^-$ be the simple zeros of $\det R_R$ and $\hat{\lambda}_1, \hat{\lambda}_2, \ldots, \hat{\lambda}_N \in C^+$ be the simple zeros of $\det R_L$, respectively, where $\lambda_l^* = \hat{\lambda}_l$ $(1 \le l \le N)$. Suppose ϱ is nonzero column vector and ϱ^* is nonzero row vector, it derives that

$$R_R(\lambda_l)\varrho_l(\lambda_l) = 0, \quad \varrho_l^*(\lambda_l^*)R_L(\lambda_l^*) = 0, \tag{3.3}$$

what is more, the eigenvectors satisfy the relation

$$\varrho_l^* = \varrho_l^{\ddagger}, \quad 1 \le l \le N, \tag{3.4}$$

$$\begin{cases} R_R(\lambda_l)\left(\dfrac{\partial \varrho_l}{\partial t} - i\lambda_l \sigma_3 \varrho_l\right) = 0, \\ R_R(\lambda_l)\left(\dfrac{\partial \varrho_l}{\partial z} - 4i\lambda_l^3 \sigma_3 \varrho_l\right) = 0. \end{cases} \tag{3.5}$$

Through substituting (3.5) into the Lax pair (2.3), we have

$$\varrho_l = e^{i\lambda_l t\sigma_3 + 4i\lambda_l^3 z\sigma_3}\varrho_{l,0}, \quad 1 \le l \le N, \tag{3.6}$$

where $\varrho_{l,0}$ is independent of the variables t, z. Considering the relation (3.4), thus we have

$$\varrho_l^* = \varrho_{l,0}^{\ddagger} e^{-i\lambda_l^* t\sigma_3 - 4i(\lambda_l^*)^3 z\sigma_3}, \quad 1 \le l \le N. \tag{3.7}$$

Therefore, the Riemann–Hilbert problem (2.21) has the solutions

$$\begin{cases} R_R(\lambda) = I - \displaystyle\sum_{m=1}^{N}\sum_{n=1}^{N} \frac{\varrho_m \varrho_n^* (M^{-1})_{mn}}{\lambda - \lambda_n^*}, \\ R_L(\lambda) = I + \displaystyle\sum_{m=1}^{N}\sum_{n=1}^{N} \frac{\varrho_m \varrho_n^* (M^{-1})_{mn}}{\lambda - \lambda_m}, \end{cases} \tag{3.8}$$

where M is the $N \times N$ matrix defined by

$$M = (m_{mn})_{N\times N} = \left(\frac{\varrho_m \varrho_n^*}{\lambda_n - \lambda_m^*}\right)_{N\times N}, \quad 1 \le m,\ n \le N, \tag{3.9}$$

and $(M^{-1})_{mn}$ is the (m,n)-element of the M^{-1}. The asymptotic expansion of $R_+(\lambda)$ can be expressed as

$$R_R(\lambda) = I + \frac{R_R^1}{\lambda} + \frac{R_R^2}{\lambda^2} + \frac{R_R^3}{\lambda^3} + o\left(\frac{1}{\lambda^4}\right), \quad \lambda \in C^- \to \infty, \tag{3.10}$$

and substituting (3.10) into Lax pair (2.3), we get

$$E = 2i(R_R^1)_{12}, \tag{3.11}$$

$$\lambda^2 : 4i[\sigma_3, R_R^1] = V_2, \tag{3.12}$$

$$\lambda^1 : 4i[\sigma_3, R_R^2] = V_2 R_R^1 + V_1, \tag{3.13}$$

$$\lambda^0 : 4i[\sigma_3, R_R^3] = V_2 R_R^2 + V_1 R_R^1 + V_0, \tag{3.14}$$

$$\lambda^{-1} : (R_R^1)_z = V_2 R_R^3 + V_1 R_R^2 + V_0 R_R^1 + V_{-1}, \tag{3.15}$$

$$\cdots.$$

From Eq. (3.8), we obtain

$$R_R^1 = -\sum_{m=1}^{N}\sum_{n=1}^{N} \varrho_m \varrho_n^* (M^{-1})_{mn}, \tag{3.16}$$

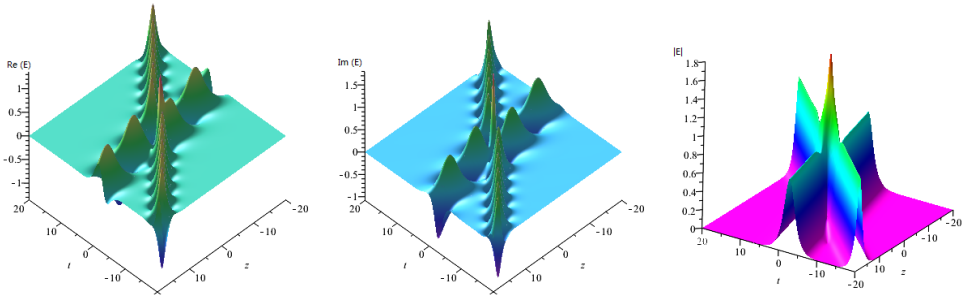


Fig. 1. (Color online) Evolution plot of $|E|$ with parameters by $\alpha_1 = \alpha_2 = 1, \beta_1 = \beta_2 = i, \lambda_1 = -\frac{1}{2} + \frac{1}{2}i, \lambda_2 = -\frac{3}{10} + \frac{2}{3}i$.

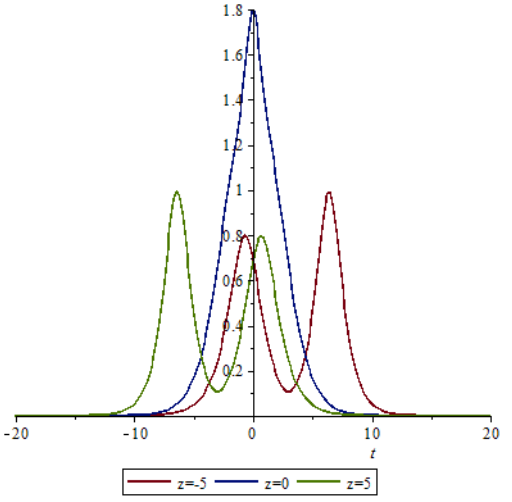


Fig. 2. (Color online) The soliton of $|E|$ along t-axis with different z.

then, the N-soliton solutions to the HMB equation (1.1) are given by

$$\begin{cases} E = -2i \sum_{m=1}^{N} \sum_{n=1}^{N} \varrho_m \varrho_n^* (M^{-1})_{mn}, \\ p = \frac{1}{2} E_z - i E_{tt} - i|E|^2 E + \frac{1}{2} E_{ttt} + 3|E|^2 E_t, \\ \eta = -E^* p - E p^*. \end{cases} \tag{3.17}$$

Then, according to the formula (3.17) established above with $N = 2$, $E(t, z)$ can be expressed as

$$E(t,z) = \frac{-2i}{m_{11}m_{22} - m_{12}m_{21}} (\alpha_1 \beta_2^* m_{12} e^{\xi_2^* - \xi_1} - \alpha_1 \beta_1^* m_{22} e^{\xi_1^* - \xi_1} + \alpha_2 \beta_1^* m_{21} e^{\xi_1^* - \xi_2} - \alpha_2 \beta_2^* m_{11} e^{\xi_2^* - \xi_2}),$$

where

$$m_{11} = \frac{1}{\lambda_1 - \lambda_1^*}(|\alpha_1|^2 e^{-\xi_1^* - \xi_1} + |\beta_1|^2 e^{\xi_1^* + \xi_1}), \tag{3.18}$$

$$m_{12} = \frac{1}{\lambda_2 - \lambda_1^*}(\alpha_1^* \alpha_2 e^{-\xi_1^* - \xi_2} + \beta_1^* \beta_2 e^{\xi_1^* + \xi_2}), \tag{3.19}$$

$$m_{21} = \frac{1}{\lambda_1 - \lambda_2^*}(\alpha_2^* \alpha_1 e^{-\xi_2^* - \xi_1} + \beta_2^* \beta_1 e^{\xi_2^* + \xi_1}), \tag{3.20}$$

$$m_{22} = \frac{1}{\lambda_2 - \lambda_2^*}(|\alpha_2|^2 e^{-\xi_2^* - \xi_2} + |\beta_2|^2 e^{\xi_2^* + \xi_2}), \tag{3.21}$$

and $\xi_1 = i\lambda_1 t + 2i\lambda_1^3 z$, $\xi_2 = i\lambda_2 t + 2i\lambda_2^3 z$. Then, some three-dimensional plots are presented shown in Fig. 1, and the solitons along the t axis with different variables are presented, as shown in Fig. 2.

4. Conclusion and Discussion

In this paper, we construct N-soliton solutions to the HMB equation via Riemann–Hilbert method. In fact, the HMB equation is a kind of important equation in light communication, which can be equivalent to nonlinear Schrödinger–Maxwell–Bloch (NLS-MB) equation under a certain constraint, so it is still very important to continue to study it. Furthermore, using Riemann–Hilbert method to solve nonlocal integrable equations and integrable hierarchies is also a very meaningful work, which not only provides a method to solve integrable equations but also promotes the development of integrable systems. In the following work, we will focus on the long-time asymptotics of integrable equations.

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